

because the prior distribution depends on the intended sample size.⁷ There could nevertheless be a pragmatic argument for tailoring the definition of the peri-null to the resolving power of an experiment.

6 Concluding Comments

The objection that “the null hypothesis is never true” may be countered by abandoning the point-null hypothesis in favor of a peri-null hypothesis. For moderate sample sizes and relatively narrow peri-nulls, this change leaves the Bayes factor relatively unaffected. For large sample sizes, however, the change exerts a profound influence and causes the Bayes factor to be inconsistent, with a limiting value given by the ratio of prior ordinates evaluated at the maximum likelihood estimate (cf. [Jeffreys, 1961](#), p. 367 and [Morey and Rouder, 2011](#), pp. 411-412). Hence, we believe that as far as Bayes factors are concerned, there is much to lose and little to gain from adopting a peri-null hypothesis in lieu of a point-null hypothesis. Here we also derived the asymptotic sampling distribution of the peri-null Bayes factor and show that its limiting value is essentially an upper bound under the alternative and a lower bound under the null. The asymptotic distributions also provide insights to typical values of the peri-null Bayes factor at a finite n . Inconsistency may not trouble subjective Bayesians: if the peri-null hypothesis truly reflects the belief of a subjective sceptic, and the alternative hypothesis truly reflects the belief of a subjective proponent, then the Bayes factor provides the relative predictive success for the sceptic versus the proponent, and it is irrelevant whether or not this relative success is bounded. Objective Bayesians, however, develop and apply procedures that meet various desiderata (e.g., [Bayarri et al., 2012](#); [Consonni et al., 2018](#)), with consistency a prominent example. As indicated above, the desire for consistency was the primary motivation for the development of the Bayesian hypothesis test ([Wrinch and Jeffreys, 1921](#)). For objective Bayesians then, it appears the point-null hypothesis is more than just a mathematically convenient approximation to the peri-null hypothesis ([Jeffreys, 1961](#), p. 367). The peri-point mixture model (consistent only under the point-null hypothesis) and the shrinking peri-point model (incoherent because the prior width depends on sample size) may provide acceptable compromise solutions.

Regardless of one’s opinion on the importance of consistency, it is evident that seemingly inconsequential changes in prior specification may asymptotically yield fundamentally different results. Researchers who entertain the use of peri-null hypotheses should be aware of the asymptotic consequences; in addition, it generally appears prudent to apply several tests and establish that the conclusions are relatively robust.

⁷ We term a Bayes factor incoherent if the result depends on whether the data are analyzed all at once or batch by batch.

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